

Math 131A: Analysis

Discussion 7: Series Convergence and Limits of Functions

1 Convergence of Infinite Series

Ratio Test: Let $\sum a_n$ be a series of nonzero terms. The series

- (i) converges absolutely if $\limsup |a_{n+1}/a_n| < 1$
- (ii) diverges if $\liminf |a_{n+1}/a_n| > 1$
- (iii) otherwise, $\liminf |a_{n+1}/a_n| \leq 1 \leq \limsup |a_{n+1}/a_n|$ and the test gives no information

Alternating Series Test: Let (a_n) be a decreasing sequence that converges to 0. Then, the series

$$\sum_{n=0}^{\infty} (-1)^n a_n$$

converges.

p-Series Test: $\sum_{n=0}^{\infty} \frac{1}{n^p}$ converges if and only if $p > 1$.

1. Which of the following series converge?

(a)

$$\sum_{n=2}^{\infty} \left(-\frac{1}{3}\right)^n$$

(b)

$$\frac{1}{n^2 + 1}$$

(c)

$$\sum \frac{n!}{n^n}$$

(d)

$$\sum \frac{n^4}{2^n}$$

(e)

$$\sum \frac{2^n}{n!}$$

(f)

$$\sum \frac{(-1)^n n!}{2^n}$$

2. (a) Give an example of a divergent series $\sum a_n$ for which $\sum a_n^2$ converges
(b) Give an example of a convergent series $\sum a_n$ for which $\sum a_n^2$ diverges
3. Find a divergent alternating series $\sum (-1)^n a_n$ such that $a_n \geq 0$ for all $n \in \mathbb{N}$ and $\lim_{n \rightarrow \infty} a_n = 0$.
4. Find a series that converges via the Root Test, but for which the Ratio Test is inconclusive.

2 Limits of Functions

Definition: Let $A \subset \mathbb{R}$. $c \in \mathbb{R}$ is a *limit point* of A if there exists a sequence $(a_n) \subset A - \{c\}$ such that

$$\lim_{n \rightarrow \infty} a_n = c.$$

A set is *closed* if it contains all its limit points

Definition: Let $A \subset \mathbb{R}$, and $c \in A$ a limit point of A . Then, $\lim_{x \rightarrow c} f(x) = L$ if for every sequence $a_n \rightarrow c$ converging to c , we have $\lim_{n \rightarrow \infty} f(a_n) = L$.

Definition: The *right hand limit* of $f(x)$ at c is $\lim_{x \rightarrow c^+} f(x) = R$, and exists if for every sequence $a_n > c$ converging to c , $\lim_{n \rightarrow \infty} f(a_n) = R$.

The *left hand limit* of $f(x)$ at c is $\lim_{x \rightarrow c^-} f(x) = L$, and exists if for every sequence $a_n < c$ converging to c , $\lim_{n \rightarrow \infty} f(a_n) = L$.

1. Is $[0, 1] \cap \mathbb{Q}$ closed in the rational numbers? What about the real numbers?
2. Is the set of irrational numbers closed in $\mathbb{R}$?
3. For the following functions, what are the right and left limits at 0? What are the limits at $\pm\infty$?

(i) $f(x) = \frac{x^3}{|x|}$

(ii) $f(x) = \frac{\sin x}{x}$

(iii) $f(x) = \sin \frac{1}{x}$

4. Let $A \subset \mathbb{R}$. Let $f : A \rightarrow \mathbb{R}$ and $c \in \mathbb{R}$ a limit point of A . Prove the following:

$$\lim_{x \rightarrow c} f(x) = L \iff \lim_{x \rightarrow c^+} f(x) = L = \lim_{x \rightarrow c^-} f(x)$$